

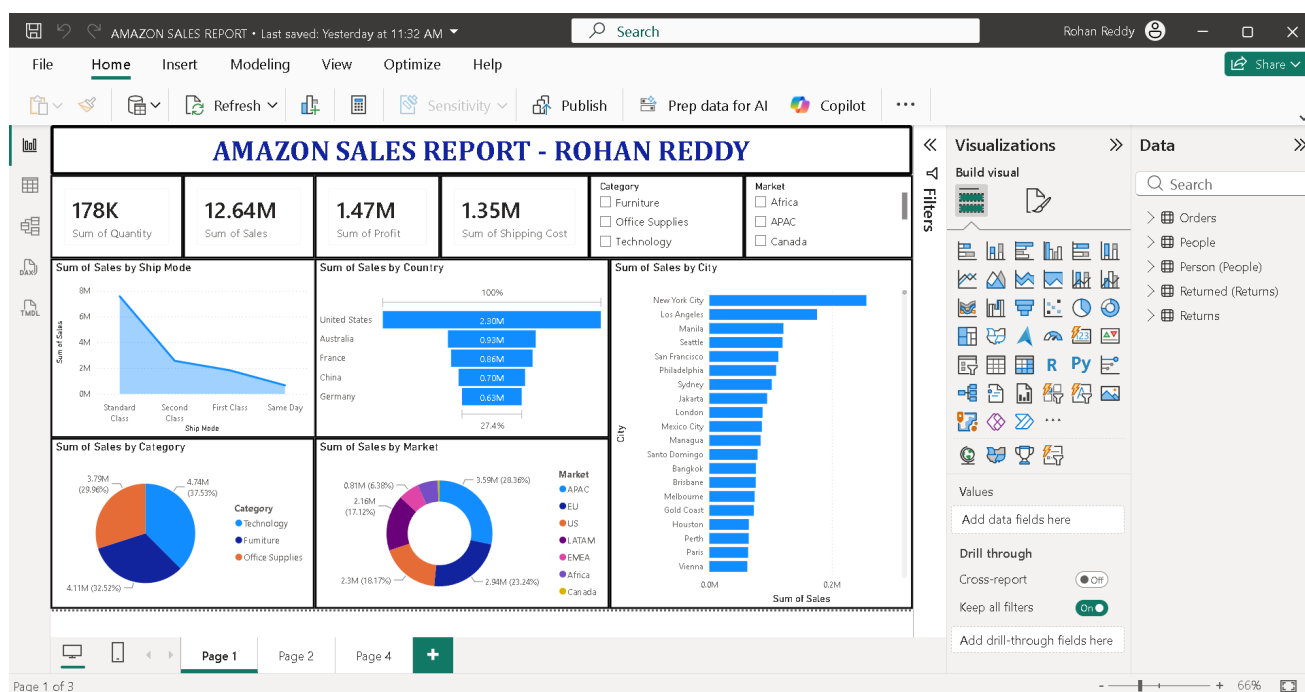
E-Commerce Sales Analytics — Executive Summary Report

Data Analytics Internship | Prepared by Rohan Reddy | Offer ID: APSPL2645037

Executive Summary

This report summarizes a full-cycle data analytics project conducted on a global e-commerce sales dataset spanning 51,290 orders across 2011–2014. The project covered data cleaning and exploratory analysis, SQL-based business querying, interactive dashboarding, and advanced statistical and predictive modeling. The dataset generated **\$12.64M in total sales** and **\$1.47M in total profit** across **25,035 unique orders** and **1,590 customers**, with an average profit margin of **11.6%**. Analysis reveals consistent year-over-year sales growth, strong seasonal patterns, distinct customer segments, and a clear link between discounting behavior and profitability — insights that directly inform pricing, marketing, and customer retention strategy.

Interactive Dashboard (Power BI)



Live dashboard: KPI overview, sales by ship mode, country, city, category, and market — with interactive category and market filters.

Top 5 Insights

1. Discounting directly erodes profitability

Orders discounted above ~30% consistently turn unprofitable on average; at 85% discount, average profit per order falls to -\$1,534. A Linear Regression model confirms Discount as the single strongest predictor of Sales value ($R^2 = 0.65$).

2. Sales show consistent growth with strong seasonality

Time series decomposition confirms a steady upward trend (~\$200K to ~\$350K monthly sales, 2011–2014) with a repeating annual pattern — dips in Jan/Feb, peaks toward year-end holiday season.

3. Four distinct customer segments exist, led by a small high-value group

K-Means clustering identified a 222-customer 'VIP' segment generating the highest average spend and profit (~\$3,161/customer) — disproportionate value from a small base.

4. Technology is the top-performing category; Central region is most profitable

Technology leads in both total sales (\$4.74M) and profit (\$663K). Regionally, Central generates the highest total profit (~\$311K), while Canada and Southeast Asia are barely profitable.

5. Order priority and profitability are statistically independent

A t-test found no significant profit difference between Consumer and Corporate segments ($p = 0.848$); similarly, 'Critical' priority orders are not meaningfully more profitable than 'Low' priority ones — suggesting priority labels don't currently reflect profit impact.

Business Recommendations

#	Recommendation	Rationale
1	Cap discounts at ~20–25% or require managerial approval beyond that threshold.	Profit turns negative past ~30% discount; capping protects margin on high-volume categories.
2	Build a dedicated retention program for the ~222 VIP customers identified via clustering.	This segment drives outsized profit per customer; losing them would disproportionately hurt profit.
3	Align inventory and staffing with the confirmed year-end seasonal demand peak.	Time series decomposition shows a reliable, repeating Q4 sales surge — plan ahead of it.